Privacy-Preserving Federated Learning

for Genomic Stage Stratification in Lung Adenocarcinoma

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Abstract

Federated learning offers a way to develop predictive models across distributed clinical datasets without transferring patient records to a central training server. This paper presents Privacy-LUAD, a web-based research dashboard developed to support federated genomic classification of lung adenocarcinoma (LUAD) into early (Stages I–II) and advanced (Stages III–IV) disease groups. The system builds on somatic mutation and clinical data collected from seven LUAD cohorts and uses a bidirectional long short-term memory (BiLSTM) model trained within a simulated three-client federated environment. Federated averaging is used to combine locally trained model parameters. A Flask-based interface provides patient-level predictions, model performance visualizations, and research reporting functions. The implementation brings the trained model and experimental outputs into a single interface, allowing users to examine predictions without working directly with the underlying code. The dashboard is intended as a research prototype for LUAD stage-group classification rather than a tool for cancer detection or clinical diagnosis.

Keywords: Federated Learning, Lung Adenocarcinoma, BiLSTM, Genomic Classification, Research Dashboard

1. Introduction

The stage of lung adenocarcinoma (LUAD) at diagnosis is an important consideration when planning treatment and assessing patient prognosis. In this study, we consider two groups: early disease, covering Stages I and II, and advanced disease, covering Stages III and IV. This grouping follows the broader clinical distinction between earlier and more advanced cancer, although staging itself depends on imaging, pathological examination, and established TNM criteria [1]. Our interest is in whether the genomic records of patients in these two groups contain patterns that a computational model can learn. Somatic mutations provide a useful starting point because they describe changes found in tumour DNA. Patients with LUAD do not necessarily carry the same mutations, and those differences may contain information relevant to disease stage. We therefore examine mutation profiles alongside available clinical attributes, while recognising that genomic findings alone cannot establish a patient's clinical stage.

Previous studies have used machine learning to investigate cancer data and explore relationships between genetic features and clinical outcomes [2]. One difficulty is that the data needed for such studies often come from different hospitals or research centres. Combining these records in one place may not be possible because of patient privacy requirements and restrictions on data sharing. Federated learning addresses part of this problem by allowing separate institutions to train a shared model without sending their raw patient records to a central server [3]. Each participating site works with its own data and contributes model updates to the training process. However, training a model is only one part of a research system. Researchers also need a way to examine its predictions, understand where it makes mistakes, and review the results without repeatedly running code. This is especially useful when the model is being tested rather than used in routine clinical practice.

Our work builds on the existing Privacy-LUAD genomic classification framework and focuses on making its federated model accessible through a web-based research dashboard. The underlying study brings together somatic mutation and clinical records from seven LUAD cohorts. For federated training, the labelled data were divided among three simulated hospital clients, each using a bidirectional long short-term memory (BiLSTM) model. Their model updates were combined using Federated Averaging [4]. We developed a Flask interface that allows users to enter supported genomic and clinical inputs and view the estimated probabilities of early or advanced LUAD. The same portal provides access to performance measures, ROC curves, confusion matrices, and saved experimental reports. The main contribution is therefore not a new learning algorithm, but a

working research environment that connects the trained model with its prediction and evaluation tools. At this stage, Privacy-LUAD remains an experimental system. It has not been validated for independent clinical decisions or cancer detection.

2. Methodology

2.1 Dataset

The Privacy-LUAD framework was developed using genomic and clinical data from seven lung adenocarcinoma (LUAD) cohorts available through cBioPortal, including TCGA PanCancer Atlas 2018, MSK 2017, MSKCC 2020, MSKCC 2023, MSK NPJ Precision Oncology 2021, NCI 2022, and OncoSG 2020. The integrated dataset contained 4,998 patients, of whom 3,558 had available clinical stage labels. These patients were grouped into early-stage (Stages I–II) and advanced-stage (Stages III–IV) LUAD.

An initial list of 96 LUAD-associated driver genes was obtained from IntOGen. Transcript-specific coding DNA sequences (CDS) were retrieved from Ensembl and used as reference sequences for mutation reconstruction. Somatic mutation information from the cBioPortal cohorts was mapped to the corresponding reference sequences to generate 301-base-pair mutation-centred DNA windows. During sequence validation, genes with unresolved transcript or coordinate mismatches were excluded, leaving 89 genes for the final analysis. A total of 17,992 reference-verified mutation windows were reconstructed.

For federated model development, the 3,558 labelled patients were divided into a training pool of 2,846 patients and a held-out test set of 712 patients. The training pool was distributed across three simulated hospital clients containing 949, 949, and 948 patients, respectively. Each client maintained a separate local training and validation partition for federated learning.

2.2 Workflow/Pipeline

The Privacy-LUAD workflow was designed to simulate collaborative model training across three hospitals without requiring patient-level data to be shared between them. Following genomic preprocessing and stage labelling, 3,558 labelled patients were divided into a federated training pool of 2,846 patients and an independent test set of 712 patients.

The federated training pool was distributed across three simulated hospital clients. Hospital 1 contained 949 patients, Hospital 2 contained 949 patients, and Hospital 3 contained 948 patients. Each hospital used an 80:20 local training and validation split. Accordingly, Hospitals 1 and 2 each contained 759 training and 190 validation patients, while Hospital 3 contained 758 training and 190 validation patients.

Each simulated hospital trained a Bidirectional Long Short-Term Memory (BiLSTM) model using its locally assigned patient data. After local training, the model parameters were transmitted to a central aggregation server, where Federated Averaging (FedAvg) [4] was applied to construct an updated global model. The global model was then redistributed to the clients for further training. The study examined federated configurations with three and five communication rounds.

Following training, the global model was evaluated using the independent test set of 712 patients. Its performance was compared with the centralized learning approach using classification metrics, including accuracy and ROC-AUC. The trained model and preprocessing components were subsequently integrated into a Flask-based research dashboard. The interface allows users to enter supported genomic and clinical information, obtain early-stage or advanced-stage LUAD prediction probabilities, and review stored evaluation results. Figure 1 presents the overall framework.

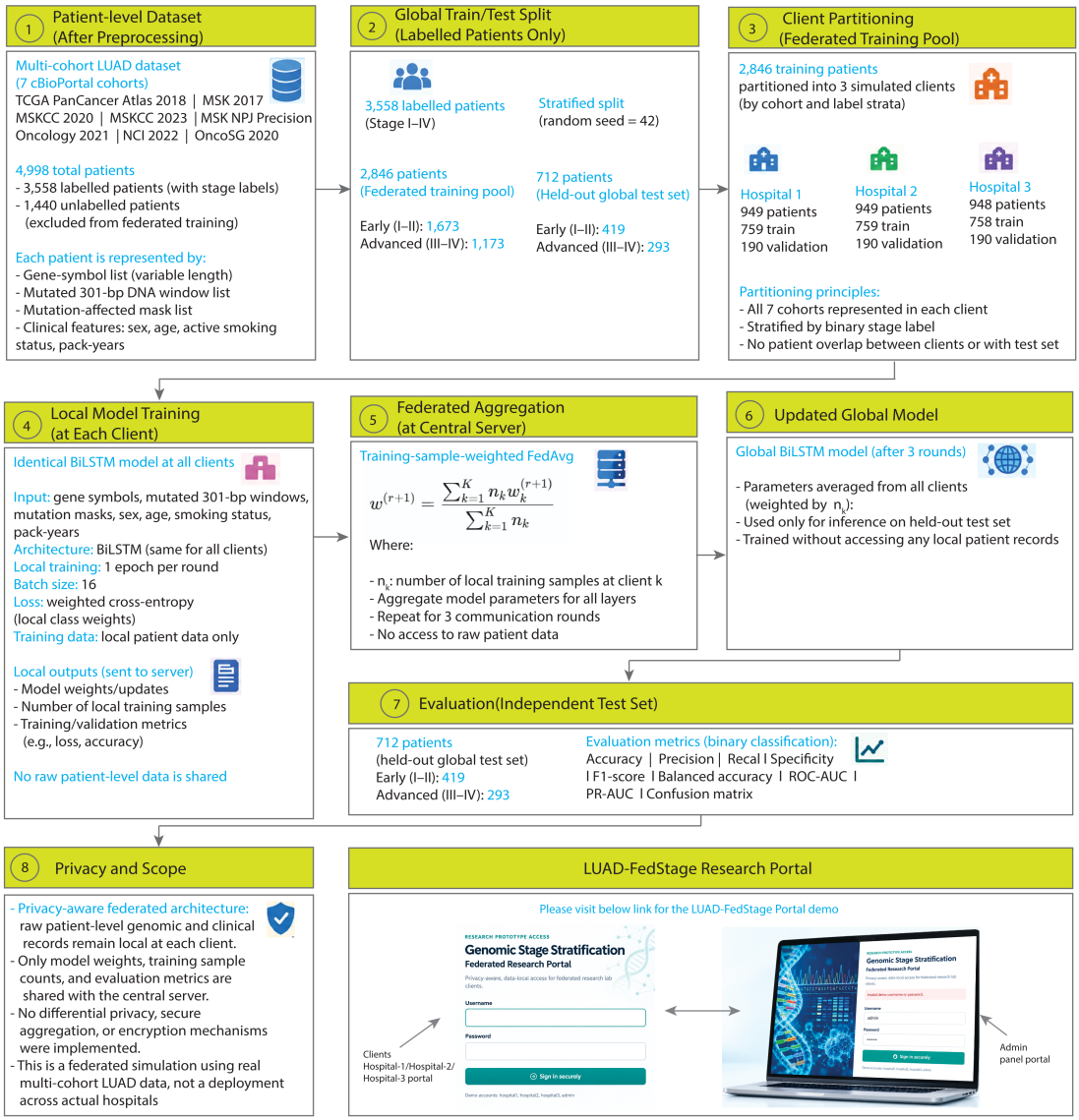


Figure 1: Privacy-LUAD federated learning workflow showing three simulated hospital clients, local BiLSTM training, FedAvg aggregation, global model evaluation, and integration with the Flask-based research dashboard.

3. Experimental Results

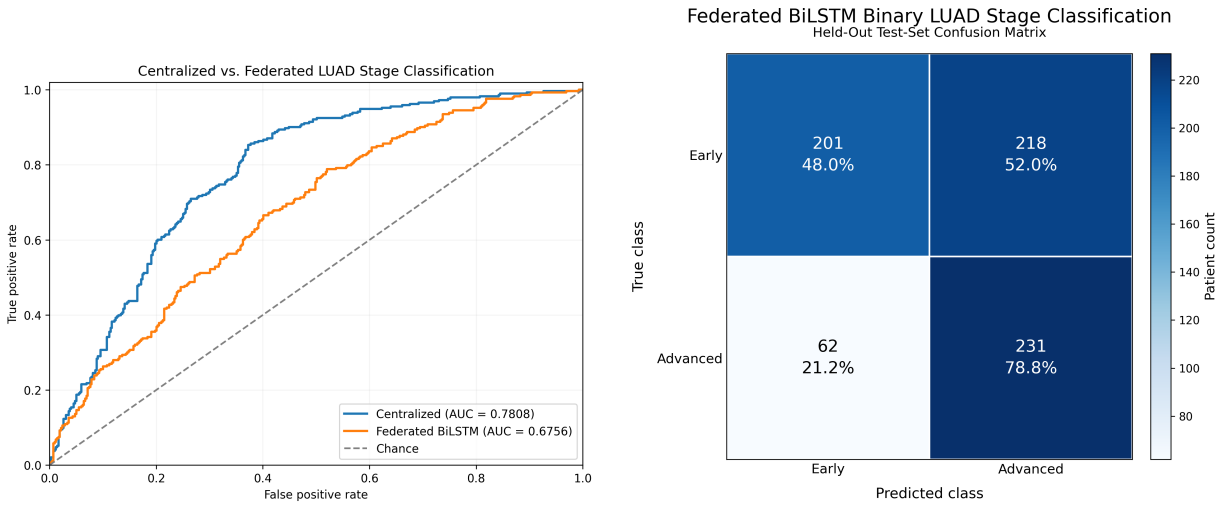
The Privacy-LUAD framework was evaluated using an independent test set of 712 patients to assess its ability to distinguish early-stage (I–II) from advanced-stage (III–IV) lung adenocarcinoma. The federated BiLSTM model achieved an accuracy of 60.67% and a ROC-AUC of 0.6756, while the centralized model achieved 70.22% accuracy and a ROC-AUC of 0.7808 on the matched test set. The results showed better predictive performance for centralized training under the evaluated experimental conditions.

Table 1: Performance comparison on the matched 712-patient test set.

Model	Acc. (%)	F1	AUC	Sens. (%)	Spec. (%)	PR-AUC
Centralized	70.22	0.7072	0.7808	87.37	58.23	0.6599
Federated	60.67	0.6226	0.6756	78.84	47.97	0.5753

The centralized model achieved 9.55 percentage points higher accuracy than the federated model. Paired bootstrap analysis with 10,000 resamples produced a 95% confidence interval of 6.04–13.06 percentage points for this difference. The ROC-AUC difference was 0.1052, with a 95% confidence interval of 0.0738–0.1375. DeLong's test also indicated a statistically significant difference in ROC-AUC ($p = 8.64 \times 10^{-11}$). These findings show that centralized training achieved stronger predictive performance under the evaluated conditions.

The discrimination performance of the centralized and federated BiLSTM models, together with the class-specific prediction outcomes of the federated model on the matched 712-patient test set, are presented in Fig. 2.



(a) ROC comparison

(b) Federated confusion matrix

Figure 2: Evaluation of centralized and federated BiLSTM models on the matched 712-patient test set: (a) ROC curve comparison and (b) federated confusion matrix.

4. Discussion

The results showed a clear difference between centralized and federated training. The centralized model performed better on the held-out test set, particularly in terms of ROC-AUC. However, the federated experiment still showed that a BiLSTM model could be trained across three simulated hospital clients without moving their patient-level training records to a central location. The lower performance may be related to differences in the data assigned to each client, the limited number of communication rounds, or the difficulty of learning stage-related patterns from genomic information. Further experiments would be needed to determine how much each factor contributed to the observed difference.

For this experiment, the three hospital clients were simulated using patient records from the same integrated LUAD dataset. We did not train the model using data collected separately from three hospitals. This is an important difference, as hospitals may have different patient populations, mutation profiles, and clinical recording practices. Differences in local datasets can affect federated learning [5], while previous studies have also examined its use in healthcare settings [3, 6]. Our results show that the federated model could distinguish between early and advanced LUAD to some extent, but its predictive accuracy remains too limited for clinical staging. The current Privacy-LUAD system is therefore intended for research and demonstration rather than clinical decision-making.

4.1 Limitations

The study has several limitations. Since the dataset included only LUAD patients and no matched non-cancer controls, the model was designed for stage classification rather than cancer detection. The three hospital clients were simulated, and the main federated experiment involved only three communication rounds, leaving model convergence uncertain. Although paired bootstrap confidence intervals and a correlated DeLong test supported the centralized and federated comparison, the initial BiLSTM checkpoint configuration could not be fully reconstructed from the retained records. The held-out test set had also been evaluated before continuation training, but it was not used for model updates, hyperparameter tuning, stopping decisions, or checkpoint selection. In addition, four-class Stage I–IV classification remained challenging, as none of the evaluated models exceeded the Stage I majority-class accuracy baseline.

4.2 Future Research Directions

We plan to take Privacy-LUAD beyond the current prototype and gradually develop it into a system suitable for testing in hospitals. One of the first requirements is a larger collection of genomic and clinical records, including samples from people without cancer. This would allow us to study cancer detection alongside LUAD stage classification. We also want to explore whether Oxford Nanopore Technologies MinION could provide sequencing data for the system and support faster genomic analysis [9]. For the prediction model, we intend to investigate transformer architectures instead of relying only on BiLSTM, and compare their performance using the same patient-level evaluation approach. As the system develops, we will need to test federated training with more participating hospitals and examine methods such as secure aggregation and differential privacy [7, 8]. Our longer-term aim is to evaluate the complete system with clinical partners and determine what improvements are needed before hospital deployment.

5. Conclusion

This study examined LUAD stage classification using mutation-centred genomic sequences and a federated BiLSTM framework. The corrected 301-bp DNA representation improved the CNN ROC-AUC from 0.6832 in the preliminary 101-bp experiment to 0.7652. In the matched comparison, the centralized BiLSTM achieved a ROC-AUC of 0.7808, while the federated model reached 0.6756. Predicting individual Stage I–IV categories remained more difficult, with no model exceeding the majority-class baseline in raw accuracy. Although federated training did not match centralized performance, the experiment showed that stage-related genomic patterns could be learned across three simulated hospital clients without pooling their patient-level training records. These findings highlight the value of careful genomic sequence reconstruction and provide an initial basis for further research into decentralized LUAD stage classification. Independent clinical validation and additional federated experiments are still needed before the framework can be considered for clinical use.

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